Trabajo Practico 1: Transformada Z

Ejercicio 1

Encontrar la transformada Z de las siguientes señales discretas e indicar su ROC (Region Of Convergence)

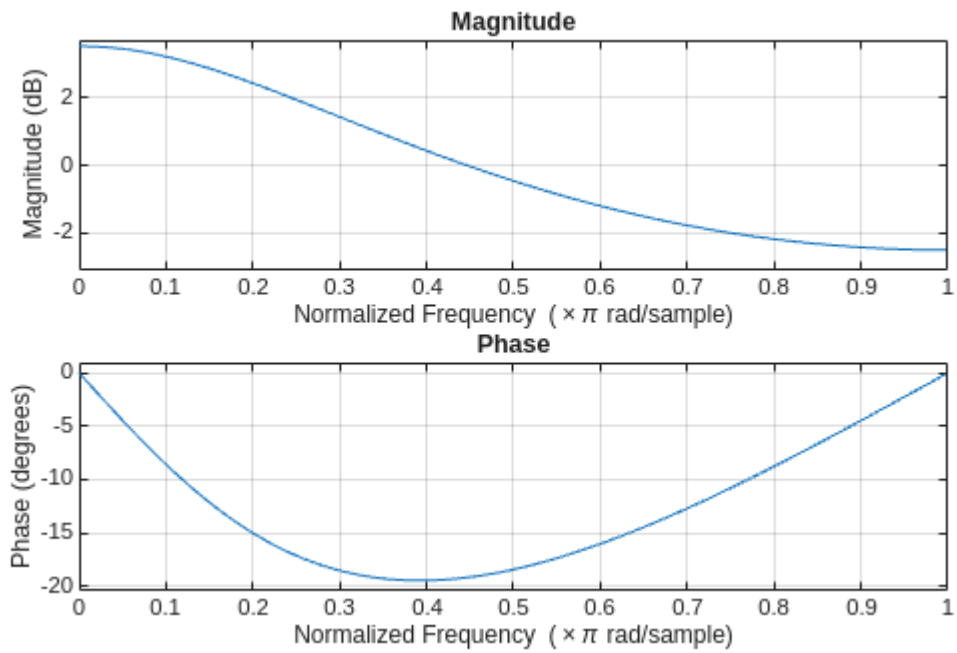
1. $x[n] = \left[\frac{1}{3}\right]^n u[n]$
2. $x[n] = -\left[\frac{1}{2}\right]^n u[-n-1]$
3. $x[n] = \left[\frac{1}{2}\right]^n u[n] - [2]^n u[-n-1]$
4. $x[n] = -\left[\frac{1}{2}\right]^n u[n] - \left[\frac{1}{2}\right]^n u[-n-1]$

1)

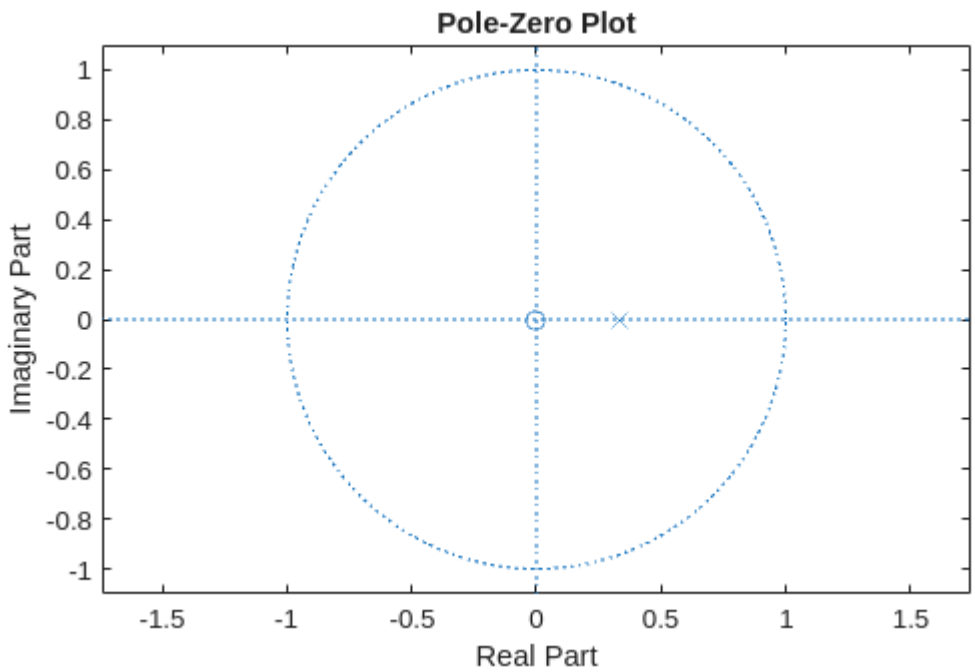
$$
\begin{aligned}
X(z) &= \sum_{n=-\infty}^{\infty} x[n] z^{-n} \\
&= \sum_{n=-\infty}^{\infty} \left[\frac{1}{3}\right]^n u[n] z^{-n} \\
&= \sum_{n=0}^{\infty} \left(\frac{1}{3} z^{-1}\right)^n \\
&= \frac{1}{1 - \frac{1}{3} z^{-1}}, \quad |z| > \frac{1}{3}
\end{aligned}
$$

Esta es una señal **causal**.

```
syms n z
x_1 = (1/3)^n;
X1 = ztrans(x_1, n, z);            % 1. Transformada simbólica
[num, den] = numden(X1);           % 2. Separar numerador y denominador
b = sym2poly(num);                 % 3. Convertir a vectores numéricos
a = sym2poly(den);
sys = tf(b, a, -1);                % 4. Crear objeto de sistema
freqz(b,a);
```




```
zplane(b, a);                    % 5. Analizar estabilidad
```



2)

$$
\begin{aligned}
Z\{x[n]\} &= \sum_{n=-\infty}^{\infty} x[n]z^{-n} \\
&= \sum_{n=-\infty}^{\infty} -\left[\frac{1}{2}\right]^n u[-n-1]z^{-n} \\
&= \sum_{n=-\infty}^{-1} -\left[\frac{1}{2}\right]^n z^{-n} \\
&= \sum_{n=-\infty}^{-1} -(2z)^{-n} \\
&= \sum_{n=1}^{\infty} -(2z)^{n} \\
&= -\frac{2z}{1-2z} \\
&= \frac{1}{1-\frac{1}{2}z^{-1}}, \qquad |z| < |\frac{1}{2}|
\end{aligned}
$$

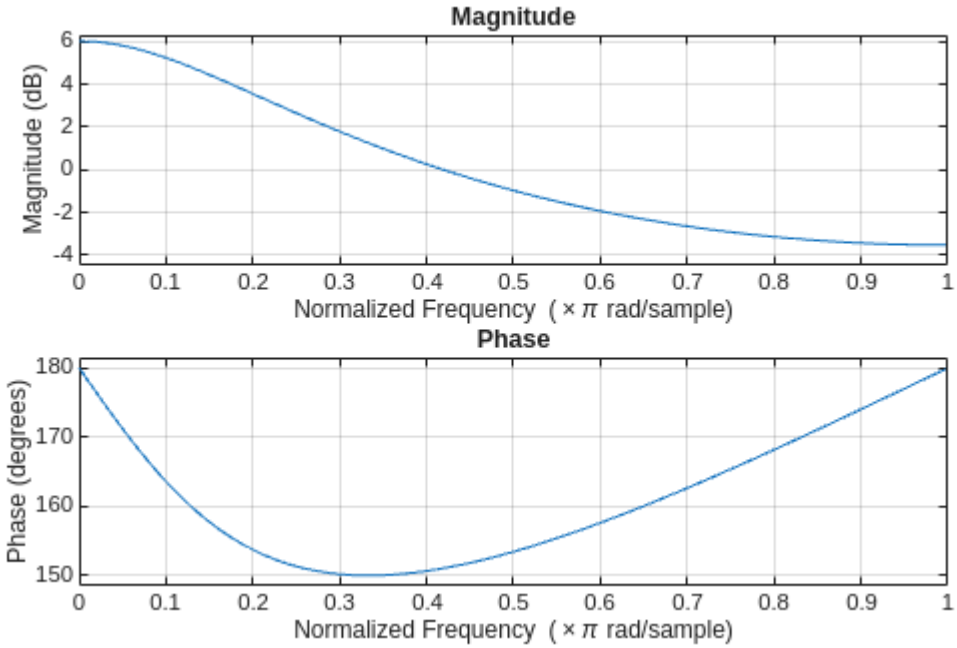
Esta es una señal **anti-causal**.

```
syms n z
x_2 = (1/2)^n;
assume(abs(z) < 0.5);
X2 = symsum(x_2 * z^(-n), n, -inf, -1);
X2 = simplify(X2)
```

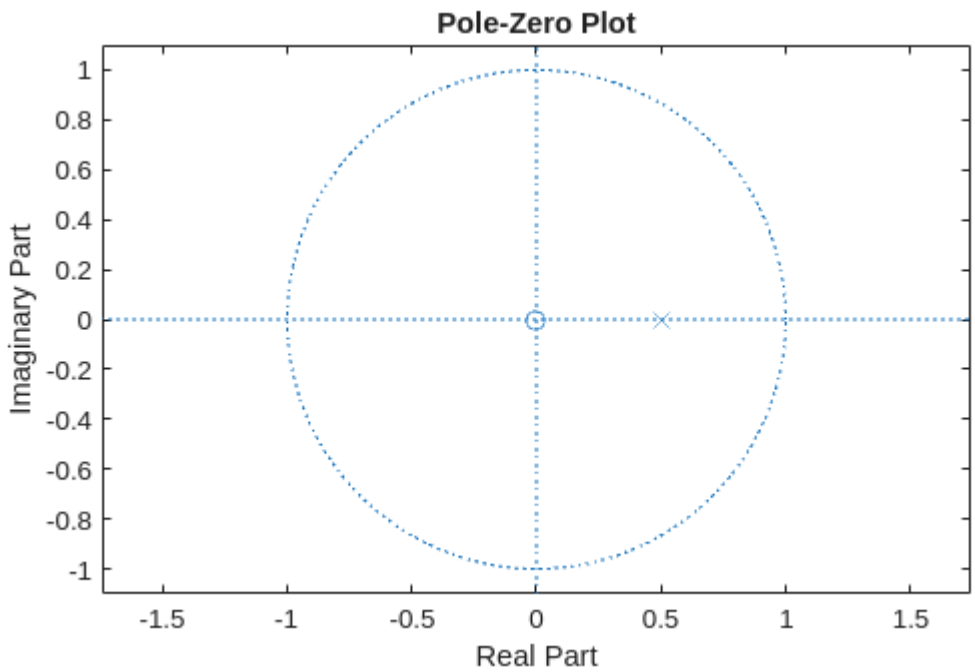
```
X2 =
```

$$-\frac{2\,z}{2\,z-1}$$

```
% Multiplicando por (-2z)/(-2z) se llega a la misma exprecion que arriba
b = [-1 0];
a = [1 -0.5];

sys = tf(b, a, -1);                % 4. Crear objeto de sistema
freqz(b,a);
```


Magnitude
Magnitude (dB)
6
4
2
0
-2
-4
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
180
170
160
150
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)

```
zplane(b,a);
```


Pole-Zero Plot
Imaginary Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1.5
-1
-0.5
0
0.5
1
1.5
Real Part

3)

$$
\begin{aligned}
Z\{x[n]\} &= Z\left\{\left[\frac{1}{2}\right]^n u[n] - [2]^n u[-n-1]\right\} \\
&= Z\left\{\left[\frac{1}{2}\right]^n u[n]\right\} + Z\left\{-[2]^n u[-n-1]\right\} \\
&= \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{1}{1-2z^{-1}}
\end{aligned}
$$

$$
= \frac{\left(1-\frac{1}{2}z^{-1}\right) + (1-2z^{-1})}{\left(1-\frac{1}{2}z^{-1}\right)(1-2z^{-1})}
$$

$$
= \frac{2-\frac{5}{2}z^{-1}}{1-\frac{5}{2}z^{-1}+z^{-2}}, \qquad \frac{1}{2} < |z| < 2
$$

```
syms n z
u1 = (1/2)^n;
U1 = ztrans(u1, n, z)
```

U1 =

$$\frac{z}{z-\frac{1}{2}}$$

```
u2 = -(2)^n;
assume(abs(z) < 2);                        % Para que devuelva expresion
cerrada (comentar para ver caso contrario)
U2 = symsum(u2 * z^(-n), n, -inf, -1)
```

U2 =

$$\frac{z}{z-2}$$

```
X3 = U1 + U2;
X3 = simplify(X3)
```

X3 =

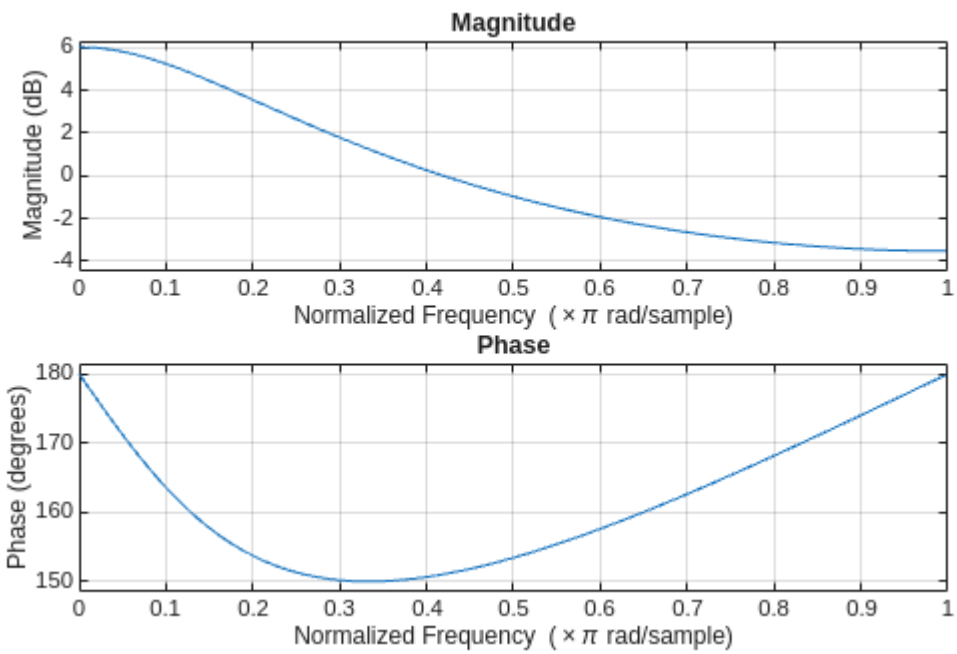
$$\frac{z}{z-2} + \frac{z}{z-\frac{1}{2}}$$

```
pretty(X3)
```

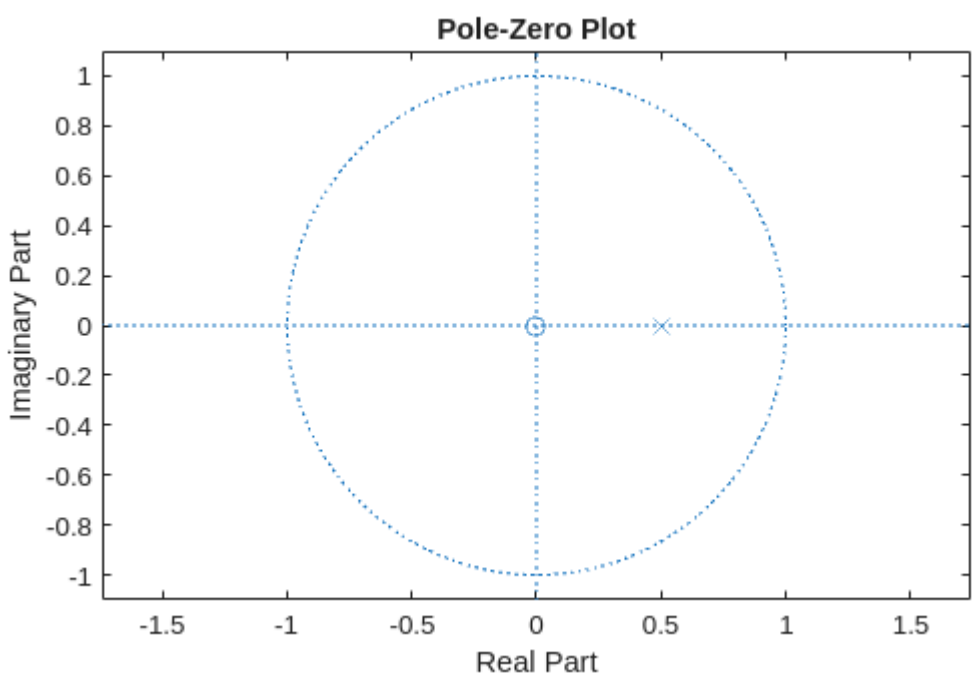
```
  z       z
----- + -----
z - 2       1
```

```
z - -
    2
```

```
% Multiplicando por (-2z)/(-2z) se llega a la misma exprecion que arriba
b = [-1 0];
a = [1 -0.5];

sys = tf(b, a, -1);                % 4. Crear objeto de sistema
freqz(b,a);
```




```
zplane(b,a);
```



4)

$$
\begin{aligned}
Z\{x[n]\} &= Z\left\{-\left[\frac{1}{2}\right]^n u[n]\right\} + Z\left\{-\left[\frac{1}{2}\right]^n u[-n-1]\right\} \\
&= -\frac{1}{1-\frac{1}{2}z^{-1}} + \frac{1}{1-\frac{1}{2}z^{-1}} \\
&= 0
\end{aligned}
$$

La señal es **no causal** y **diverge.**

Analizando el ROC de las dos señales por separado:

$$ROC_1 : \frac{1}{2} < |z|$$

$$ROC_2 : \frac{1}{2} > |z|$$

No existe numero que pueda cumplir ambos requisitos simultaneamente, ergo no existe ROC ($ROC = \varnothing$).

```
syms n z
u1 = -(1/2)^n;
U1 = ztrans(u1, n, z)
```

U1 =

$$-\frac{z}{z-\frac{1}{2}}$$

```
u2 = -(1/2)^n;
assume(abs(z) < 0.5);                        % Para que devuelva expresion
cerrada (comentar para ver caso contrario)
U2 = symsum(u2 * z^(-n), n, -inf, -1)
```

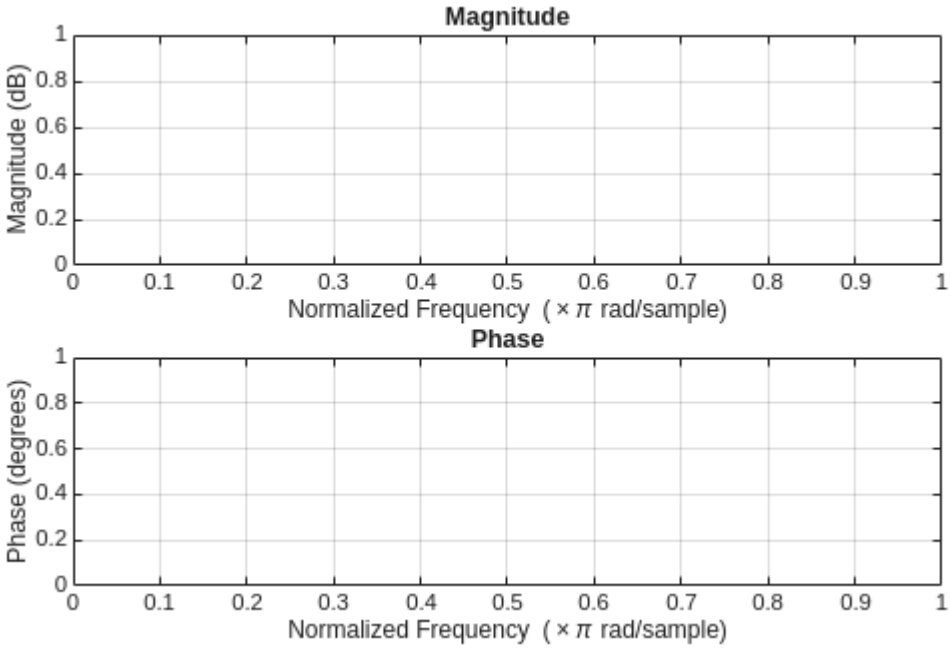
U2 =

$$\frac{2z}{2z-1}$$

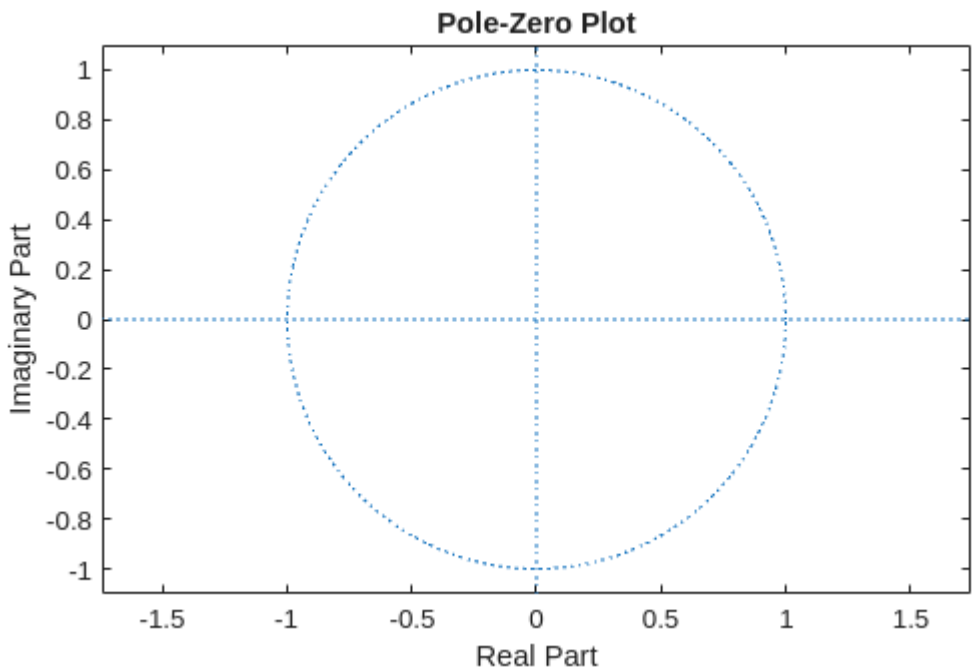
```
X4 = U1 + U2;
X4 = simplify(X3)
```

X4 = 0

```
% Multiplicando por (-2z)/(-2z) se llega a la misma exprecion que arriba
b = [0 0];
a = [1];

sys = tf(b, a, -1);                % 4. Crear objeto de sistema
freqz(b,a);
```




```
zplane(b,a);
```



Ejercicio 2

Considerando las transformadas:

$$X_1(z) \quad = 1 + z^{-1} + 3z^{-2}$$
$$X_2(z) \quad = 1 + 3z^{-2}$$

Su convolución con propiedades de transformada Z es:

$$
\begin{aligned}
X(z) &= X_1(z) \cdot X_2(z) \\
&= (1 + z^{-1} + 3z^{-2})(1 + 3z^{-2}) \\
&= (1 + 3z^{-2}) + z^{-1}(1 + 3z^{-2}) + 3z^{-2}(1 + 3z^{-2}) \\
&= 1 + z^{-1} + 6z^{-2} + 3z^{-3} + 9z^{-4}
\end{aligned}
$$

y llevandolo al dominio del tiempo:

$$x[n] = \delta[n] + \delta[n-1] + 6\delta[n-2] + 3\delta[n-3] + 9\delta[n-4]$$

Haciendo las operaciones en el dominio del tiempo:

$$
\begin{aligned}
x_1[n] &= \delta[n] + \delta[n-1] + 3\delta[n-2] \\
x_2[n] &= \delta[n] + 3\delta[n-2]
\end{aligned}
$$

$$x_1[n] = x_2[n] + \delta[n-1]$$

y realizando la convolución:

$$x_1[n] * x_2[n] = \sum_{m=-\infty}^{\infty} x_1[n-m]x_2[m]$$

Por propiedad de conmutación de la convolución, se puede hacer:

$$
\begin{aligned}
x_1[n] * x_2[n] &= \Big(x_2[n] + \delta[n-1]\Big) * x_2[n] \\
&= x_2[n] * x_2[n] + x_2[n] * \delta[n-1]
\end{aligned}
$$

Por propiedad de "identidad multiplicativa" de convolución, se tiene:

$$
\begin{aligned}
x_2[n] * \delta[n-1] &= \sum_{m=-\infty}^{\infty} x_2[n-m]\delta[m-1] \\
&= x_2[n-1] \\
&= \delta[n-1] + 3\delta[n-3]
\end{aligned}
$$

esto es así debido a que $\delta = 0 \ \forall n \neq 1$ y $\delta = 1 \iff n = 1$.

Luego, desarrollando la otra convolución:

$$
\begin{aligned}
x_2[n] * x_2[n] &= \sum_{m=-\infty}^{\infty} \Big(\delta[n-m] + 3\delta[n-2-m]\Big)\Big(\delta[m] + 3\delta[m-2]\Big) \\
&= \sum_{m=-\infty}^{\infty} \delta[n-m]\delta[m] + 3\delta[n-m]\delta[m-2] + 3\delta[n-2-m]\delta[m] + 9\delta[n-2-m]\delta[m-2] \\
&= \sum_{m=-\infty}^{\infty} \delta[n-m]\delta[m] + \sum_{m=-\infty}^{\infty} 3\delta[n-m]\delta[m-2] + \sum_{m=-\infty}^{\infty} 3\delta[n-2-m]\delta[m] + \sum_{m=-\infty}^{\infty} 9\delta[n-2-m]\delta[m-2] \\
&= \delta[n] + 3\delta[n-2] + 3\delta[n-2] + 9\delta[n-4]
\end{aligned}
$$

eso nos deja:

$x_1[n] * x_2[n] = \delta[n] + \delta[n-1] + 6\delta[n-2] + 3\delta[n-3] + 9\delta[n-4]$

lo cual coincide con el resultado de hacer el producto en el dominio Z y hacer la antitransformada (por tabla).

```
% Coeficientes de x1[n] = [z^0, z^-1, z^-2]
x1 = [1, 1, 3];
% Coeficientes de x2[n] = [z^0, z^-1, z^-2]
x2 = [1, 0, 3];

% Punto 4: Multiplicar polinomios
x_conv = conv(x1, x2);

disp('Resultado con conv:');
```

```
Resultado con conv:
```

```
disp(x_conv);
```

```
     1     1     6     3     9
```

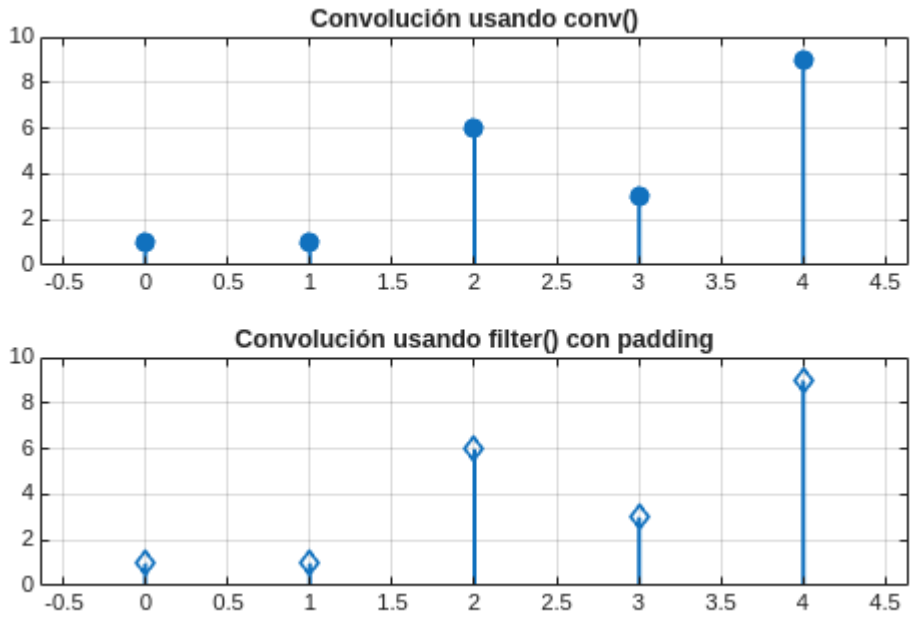
```
% Debería devolver: 1  1  6  3  9

L_total = length(x1) + length(x2) - 1;
x1_padded = [x1, zeros(1, length(x2)-1)];

% Filtramos x1_padded usando x2 como coeficientes del numerador
% ¿Por que?
x_filter = filter(x2, 1, x1_padded)
```

```
x_filter = 1×5
     1     1     6     3     9
```

```
x_wo_padding = filter(x2, 1, x1)
```

```
x_wo_padding = 1×3
     1     1     6
```

```

% Comparar resultados
n = 0:L_total-1;

figure;
subplot(2,1,1);
stem(n, x_conv, 'filled', 'LineWidth', 1.5);
title('Convolución usando conv()');
grid on;

subplot(2,1,2);
stem(n, x_filter, 'd', 'LineWidth', 1.5);
title('Convolución usando filter() con padding');
grid on;
```




```
% Verificación lógica
if isequal(x_conv, x_filter)
    disp('Verificación exitosa: Los métodos coinciden.');
else
    disp('Error: Los métodos no coinciden. Revisar el padding.');
end
```

```
Verificación exitosa: Los métodos coinciden.
```

Ejercicio 3

Encontrar el ROC de:

$$x[n] = [0.5]^n u[n] + [0.3]^n u[n] + [0.9]^n u[n]$$

Analizando por separado, tenemos:

$$Z\left\{[0.5]^n u[n]\right\} = \frac{1}{1-0.5z^{-1}}, \qquad 0.5 < |z|$$

$$Z\left\{[0.3]^n u[n]\right\} = \frac{1}{1-0.3z^{-1}}, \qquad 0.3 < |z|$$

$$Z\left\{[0.9]^n u[n]\right\} = \frac{1}{1-0.9z^{-1}}, \qquad 0.9 < |z|$$

El ROC de $x[n]$ sera: $ROC : 0.9 < |z|$.

Luego, por propiedades de transformada Z:

$$Z\left\{ ax[n] + by[n] \right\} = aZ\left\{ x[n] \right\} + bZ\left\{ y[n] \right\}$$

$$Z\left\{ x[n] \right\} = \frac{1}{1 - 0.5z^{-1}} + \frac{1}{1 - 0.3z^{-1}} + \frac{1}{1 - 0.9z^{-1}}$$

Ejercicio 4

Considerando la ecuacion en diferencias:

$$y[n] - 0.7y[n-1] = x[n]$$

- La funcion de transferencia del sistema sera:

$$\begin{aligned}
y[n] - 0.7y[n-1] &= \delta[n] \\
Z\left\{ y[n] - 0.7y[n-1] \right\} &= Z\left\{ \delta[n] \right\} \\
Y(z) - 0.7z^{-1}Y(z) &= 1 \\
(1 - 0.7z^{-1})Y(z) &= 1 \\
Y(z) &= \frac{1}{(1 - 0.7z^{-1})} \\
H(z) &= \frac{1}{(1 - 0.7z^{-1})}
\end{aligned}$$

- Para $x[n] = \delta[n]$

$$Y(z) = H(z) = \frac{1}{(1 - 0.7z^{-1})}$$

- Para $x[n] = u[n]$

$$\begin{aligned}
Y(z) &= X(z)H(z) = \frac{1}{1 - z^{-1}} \cdot \frac{1}{(1 - 0.7z^{-1})} \\
&= \frac{1}{1 - 1.7z^{-1} + 0.7z^{-2}}
\end{aligned}$$

Ejercicio 5

Considerando el sistema:

$$\begin{cases} y[n] & -0.5y[n-1] - 0.1y[n-2] - 0.2y[n-3] = x[n] \\ x[n] & = u[n] \\ y[-1] & = 1 \\ y[-2] & = 2 \\ y[-3] & = 3 \end{cases}$$

El sistema es inestable, con $ROC : 1 < |z|$.

La funcion de transferencia sera:

$$y[n] - 0.5y[n-1] - 0.1y[n-2] - 0.2y[n-3] = \delta[n]$$

$$Z\left\{ y[n] - 0.5y[n-1] - 0.1y[n-2] - 0.2y[n-3] \right\} = Z\left\{ \delta[n] \right\}$$

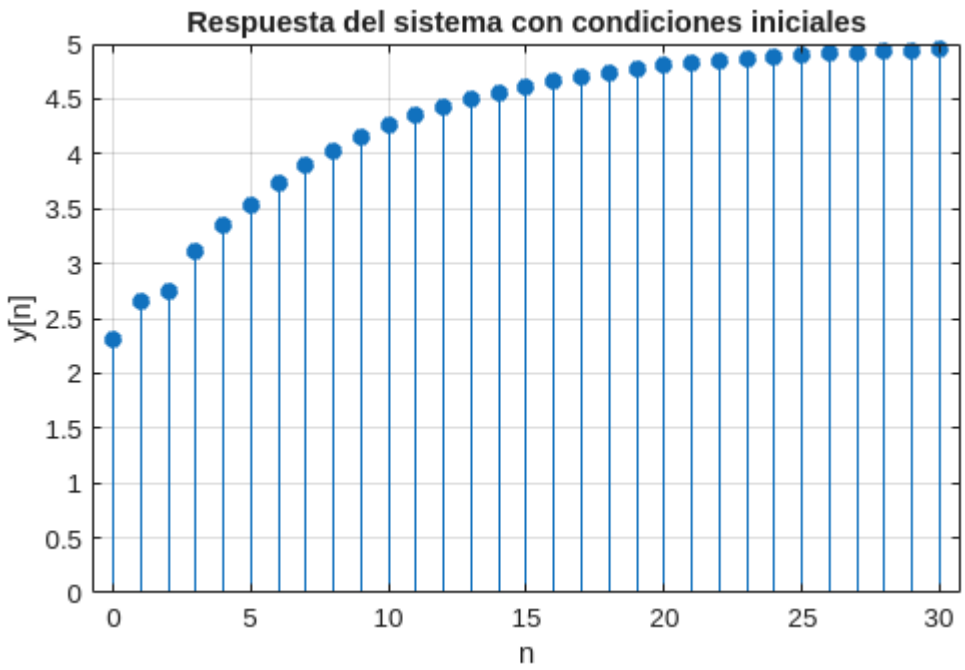
$$Y(z) - 0.5z^{-1}Y(z) - 0.1z^{-2}Y(z) - 0.2z^{-3}Y(z) = 1$$

$$(1 - 0.5z^{-1} - 0.1z^{-2} - 0.2z^{-3})Y(z) = 1$$

$$Y(z) = \frac{1}{(1 - 0.5z^{-1} - 0.1z^{-2} - 0.2z^{-3})}$$

$$H(z) = \frac{1}{(1 - 0.5z^{-1} - 0.1z^{-2} - 0.2z^{-3})}$$

```
% 1. Definir coeficientes
a = [1, -0.5, -0.1, -0.2];    % den
b = [1];                      % num

% 2. Condiciones iniciales en orden [y[-1], y[-2], y[-3]...]
y_ic = [1, 2, 3];
x_ic = [0, 0, 0];

% 3. Usar funcion filtic
zi = filtic(b, a, y_ic, x_ic);

% 4. Definir el vector de tiempo y la entrada
n = 0:30;
x_conv = ones(size(n)); % u[n]

% 5. Calcular la respuesta
y = filter(b, a, x_conv, zi);

% 6. Graficar
stem(n, y, 'filled');
grid on;
xlabel('n'); ylabel('y[n]');
title('Respuesta del sistema con condiciones iniciales');
```


Respuesta del sistema con condiciones iniciales
y[n]
n
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
0
5
10
15
20
25
30